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Assignment No. 01

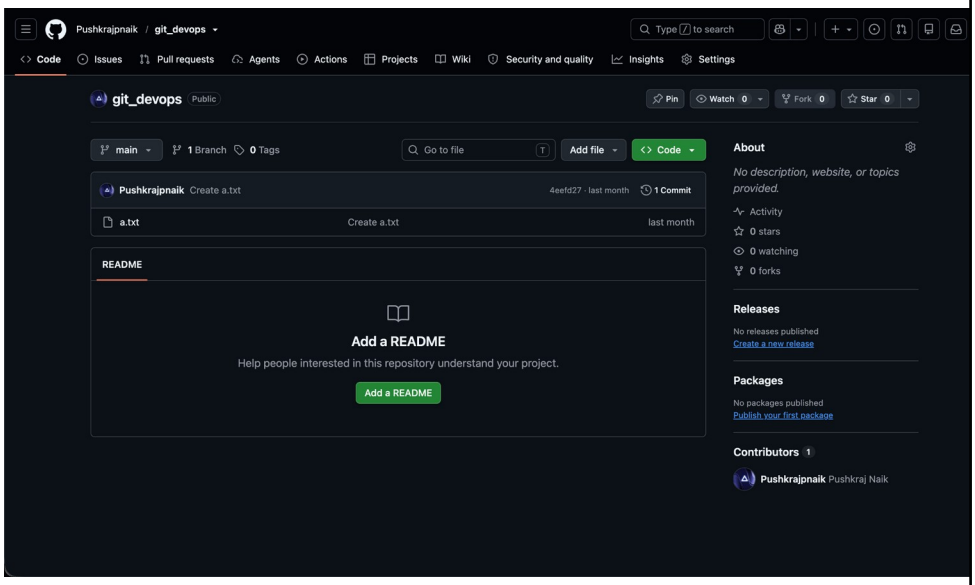
Subject:	DevOps Lab
TCODE:	TE7950
Name of Student	**Pushkraj P Naik**
PRN No.	**23070122169**
Branch	CSE, Batch (2023-27), L3
Academic Year & Semester	2025 – 2026, Semester VII
Date of Performance	16th July, 2026
Title of Assignment:	Git Setup & Basic Workflow

Theory:	Git is a distributed version control system that helps developers track changes in source code, collaborate efficiently, and maintain different versions of a project. GitHub is a cloud-based platform that hosts Git repositories and enables collaboration. In this assignment, a GitHub repository was created and cloned to the local system. A README.md file was created, staged using git add, committed with git commit, and uploaded to GitHub using git push. This demonstrates the basic Git workflow for managing and synchronizing project files between a local machine and a remote repository. **Commands Used** 1. cd D:\CCodes – Navigated to the directory where the repository would be stored. 2. git clone https://github.com/riavinod10/devops-lab.git – Cloned the remote GitHub repository to the local machine. 3. cd devops-lab – Entered the cloned repository folder. 4. echo "# DevOps Lab" > README.md – Created the README.md file with initial content.

	5. git status – Checked the repository status to verify that README.md was listed as an untracked file. 6. git add README.md – Added the README.md file to the staging area. 7. git status – Verified that the file was staged and ready to be committed. 8. git commit -m "Add initial README.md file" – Created the first commit with a descriptive commit message. 9. git status – Confirmed that the working tree was clean after the commit. 10. git push origin main – Pushed the committed changes from the local repository to the remote GitHub repository.

Implementation:

Step 1: Create a New Repository on GitHub



Empty Repository on GitHub — Created an empty repository named devops-lab on GitHub without initializing it with a README, .gitignore, or license file to practice the Git local-to-remote workflow.

Step 2: Clone the Repository

git remote set-url origin
https://github.com/Pushkrajpnaik/git_devops
git --version
git config --global user.name "Pushkraj"
git config --global user.email "pushkrajpnaik@gmail.com"

Repository Cloned- Cloned the remote GitHub repository to the local machine using git clone and navigated into the project directory.

Step 3: Create README.md

```
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % cat DevOps-Assignments/Assignment-01-Git/README.md
# DevOps Lab - Assignment 1

## Title
Git Setup and Basic Workflow

## Objective
To understand the basic Git workflow by creating a file,
checking repository status, staging changes, committing
the changes, and synchronizing with a remote repository.

## Tools Used
- Git
- GitHub
- macOS Terminal

## Procedure
1. Create a README.md file.
2. Check the repository status using git status.
3. Stage the README file using git add.
4. Commit the changes using git commit.
5. Verify the commit using git log.
6. Push the changes to the remote GitHub repository.

## Result
The basic Git workflow was demonstrated using a README file.
```

Created README.md- Created the README.md file with initial content and verified using git status that Git recognizes it as an untracked file.

Step 4: Stage the File

```
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % git add DevOps-Assignments/Assignment-01-Git/README.md
git status
On branch main
Your branch is behind 'origin/main' by 1 commit, and can be fast-forwarded.
  (use "git pull" to update your local branch)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   DevOps-Assignments/Assignment-01-Git/README.md
```

Staged README.md- Added the README.md file to the staging area using git add and confirmed it is ready to be committed.

Step 5: Commit the Changes

```
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % git commit -m "Add Assignment 1: Git basic workfl
[main f68fb87] Add Assignment 1: Git basic workflow
 1 file changed, 25 insertions(+)
 create mode 100644 DevOps-Assignments/Assignment-01-Git/README.md
```

Committed Changes- Created the first commit with a descriptive commit message and verified that the working tree is clean.

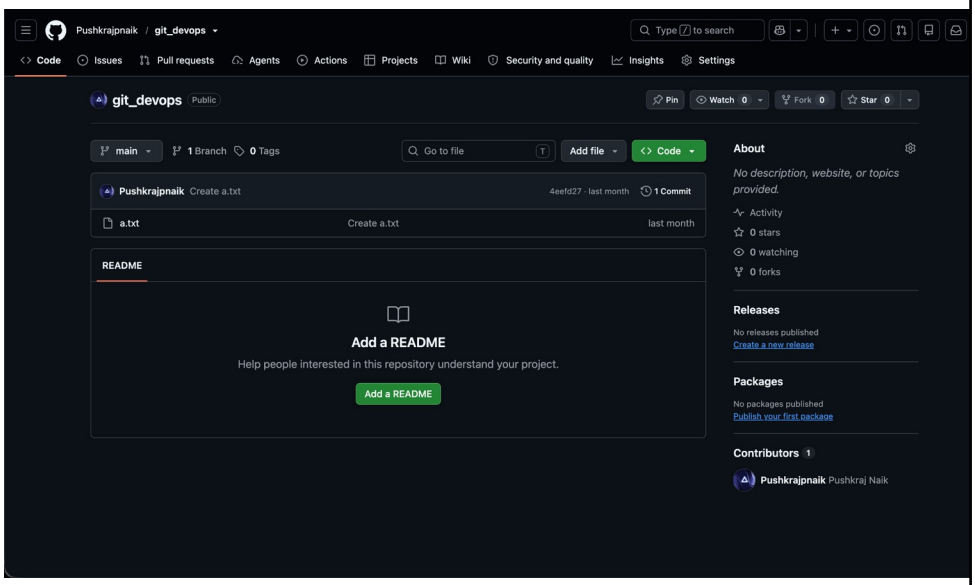
Step 6: Push to GitHub

```
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % git merge origin/main --no-edit
Merge made by the 'ort' strategy.
 23070122169/Project 1/README.md | 4 ++--
 1 file changed, 2 insertions(+), 2 deletions(-)
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % git push origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (7/7), 1.05 KiB | 1.05 MiB/s, done.
Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Pushkrajpnaik/DevOps-Lab.git
   446a2f7..ea54184  main -> main
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % git status
git log --oneline --graph --decorate -5
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
*   ea54184 (HEAD -> main, origin/main, origin/HEAD) Merge remote-tracking branch 'origin/mai
|\
| * 446a2f7 Update student name and PRN in README
* | f68fb87 Add Assignment 1: Git basic workflow
|/
* 3836972 Add DevOps lab work
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps %
```

Pushed to Remote Repository: Uploaded the local commit to the remote GitHub repository using git push.

Step 7: Verify on GitHub



Repository Verified: Verified on GitHub that the README.md file and the latest commit are visible, confirming the push was successful.

Subject Faculty Name: Prof. Arijit Dutta